



Debug Detective

Quick facilitation notes & answer key

Goal

The capstone of the subject — and of the whole K–G3 catalogue. Students read two two-move parallel scripts, diagnose WHICH is wrong and WHAT KIND of bug it is, fix it, and explain (C3.2 read/alter/describe + C3.1 write). The goal is 'meet on 3'; Bit is correct, but Pixel's 'back 3' overshoots to 1 — a wrong-number (too big) bug; the fix is 'back 1'. Naming the bug type and justifying the fix is the highest reading-and-debugging skill at Grade 2.

Ontario curriculum (Grade 2 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: a star on square 3 and two counters; trace each two-move script and see which counter sails past the star.

Run it (about 20 min)

1. I DO: state the goal — 'meet on 3.' Trace Bit: 0 forward 1 → 1, forward 2 → 3 (on the star).
2. WE DO: trace Pixel: 5 back 1 → 4, back 3 → 1 (sails past the star to 1). 'Pixel overshoot — its second number is too big.'
3. YOU DO: students name the bug (back 3 is too big — a wrong-number bug), fix it to back 1, and explain why (4 to 3 is just one back).
4. Connect: 'A detective names the bug and changes only what is wrong.'

Answer key (modelled by the run-gate)

Ex 1 — Bit: 0 → forward 1 → 1 → forward 2 → 3 (meets goal). Pixel: 5 → back 1 → 4 → back 3 → 1 (overshoots; Pixel is the buggy one).

Ex 2 — the second move 'back 3' made Pixel overshoot. It is a wrong-number bug (the number is too big), not a direction or missing-block bug.

Ex 3 — fixed move: back 1 → Pixel: 5 → back 1 → 4 → back 1 → 3. Explanation: from 4, the goal 3 is only one square back, so back 1 lands exactly on 3.

Detective note: the direction (back) was right; only the size of the second move was wrong.

Watch for & differentiate

Common error: blaming the first move (back 1 was fine) — trace move by move to see exactly where Pixel passes the goal.

Simplify: stop Pixel after each move and check against the star to spot the overshoot.

Extend: ask students to invent a different bug (wrong direction or missing block) that would also miss 3, then fix it — a fitting send-off for the whole coding series.

Success indicator: the child names Pixel's 'back 3' as a too-big number, fixes it to back 1, and explains why it now lands on 3.